

Predicting Acute Myeloid Leukemia (AML) Status from HLA Genotypes Using Classical Machine Learning

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Abstract

Acute Myeloid Leukemia (AML) is a genetically heterogeneous hematologic malignancy whose immunogenetic determinants remain incompletely understood. Human leukocyte antigen (HLA) genotypes play a central role in antigen presentation and immune surveillance, motivating the hypothesis that HLA variation may correlate with AML susceptibility. This thesis investigates whether classical machine learning models can predict AML status from high-dimensional HLA genotype data combined with demographic variables. The study emphasizes methodological rigor, including extensive preprocessing, feature engineering, linkage disequilibrium analysis, one-hot encoding, class imbalance handling, and systematic model evaluation across multiple dataset variants. Despite comprehensive experimentation, the results demonstrate that AML prediction from HLA genotypes is not feasible with the available dataset. Models trained on imbalanced data collapse to majority-class predictions, while oversampling via SMOTE improves minority recall but fails to yield clinically meaningful performance. Feature importance analyses highlight age, HLA-A, HLA-B, and DRB/DQB alleles as influential, yet these signals do not translate into robust classifiers. The findings underscore the limitations of classical machine learning in sparse, high-dimensional immunogenetic datasets and provide methodological insights for future computational immunogenetics research.

1 Introduction

Acute Myeloid Leukemia (AML) is an aggressive hematologic malignancy characterized by clonal expansion of immature myeloid precursors in the bone marrow. Its etiology involves a complex interplay of somatic mutations, cytogenetic abnormalities, and immune dysregulation. Among the immunogenetic factors implicated in leukemia biology, the human leukocyte antigen (HLA) system plays a central role in shaping immune recognition and modulating natural killer (NK) cell activity. HLA molecules present peptide antigens to T cells and influence NK cell licensing, making them plausible contributors to leukemia susceptibility and relapse dynamics [1].

The potential association between HLA alleles and AML outcomes has been explored in several studies. For example, van der Ploeg et al. demonstrated that specific HLA-A alleles, including HLA-A*24 and HLA-A*32, influence NK cell function and impact AML relapse following allogeneic hematopoietic cell transplantation [2]. Such findings motivate the hypothesis that HLA genotype patterns may contain predictive information relevant to AML status.

This thesis investigates whether classical machine learning (ML) models can predict AML status from HLA genotypes and demographic variables. The study is intentionally machine-learning focused, emphasizing data representation, encoding strategies, class imbalance, model behavior, and interpretability. Rather than aiming for clinical prediction, the goal is to rigorously evaluate the feasibility of AML classification under realistic constraints of sample size, sparsity, and allele diversity.

The central research question is therefore: *Can classical machine learning models learn a meaningful predictive signal for AML status from high-dimensional HLA genotype data?* Through systematic experimentation, this thesis demonstrates that the answer is negative for the available dataset, yet the methodological insights gained are valuable for future computational immunogenetics research.

2 Background

2.1 Acute Myeloid Leukemia

AML arises from malignant transformation of hematopoietic stem and progenitor cells. Its clinical presentation is rapid and severe, requiring immediate intervention. The disease exhibits substantial genetic heterogeneity, with recurrent mutations in genes such as FLT3, NPM1, DNMT3A, and IDH1/2. Prognosis is influenced by cytogenetic risk groups, molecular markers, and patient age. Although AML is not traditionally considered an HLA-driven disease, immune surveillance mechanisms may influence disease onset and progression [3].

2.2 The Human Leukocyte Antigen System

The HLA system comprises a highly polymorphic set of genes located on chromosome 6. Class I molecules (HLA-A, HLA-B, HLA-C) present intracellular peptides to CD8+ T cells, while class II molecules (HLA-DR, HLA-DQ, HLA-DP) present extracellular peptides to CD4+ T cells. HLA polymorphism affects antigen binding, immune recognition, and NK cell education. The extreme diversity of HLA alleles poses challenges for statistical modeling, as many alleles appear infrequently in population datasets [4, 5].

2.3 Machine Learning in Immunogenetics

Machine learning has been applied to immunogenetic problems such as peptideHLA binding prediction, transplant matching, and autoimmune disease risk modeling. However, HLA genotype data present unique challenges: high dimensionality, sparsity, linkage disequilibrium, and allele frequency imbalance. Classical ML models often struggle in such settings without large sample sizes or strong underlying signals [6]. Oversampling techniques such as SMOTE attempt to mitigate class imbalance but may introduce synthetic artifacts [7]. Interpretability methods such as SHAP provide insight into model behavior but cannot compensate for structural dataset limitations [8].

3 Dataset and Preprocessing

3.1 Raw Dataset Structure

The dataset consisted of 294 individuals with 18 columns. After removing the empty **SAL** column and the ID column, 16 usable features remained. These included age, sex, disease status, and allele values for HLA-A, HLA-B, HLA-C, DRB, and DQB loci. The disease status column was recoded such that AML = 1 and healthy = 0. The sex column was recoded such that female = 0 and male = 1.

3.2 Handling Missing Values

Several HLA loci contained missing or zero values, particularly C1, C2, DQB11, and DQB12. Zero values were treated as equivalent to missing. A total of 36 rows contained missing or zero values in C1 or C2. Based on this, three dataset variants were constructed: the original dataset with missing values retained, a complete-case dataset with all missing rows removed (245 rows), and a reduced-locus dataset with C1, C2, DQB11, and DQB12 removed prior to dropping missing rows (272 rows).

3.3 Feature Engineering

Biologically motivated features were added based on literature indicating the relevance of HLA-A*24 and HLA-A*32 to AML relapse risk [2]. Binary indicators for these alleles were created. Additional features captured homozygosity at the A, B, and C loci while excluding null or zero pairs. Linkage disequilibrium analysis revealed moderate correlations between A1A2, B1B2, and C1C2, consistent with known HLA structure [5].

3.4 Encoding

One-hot encoding was applied to all allele columns except DRB3, DRB4, DRB5, age, sex, and status. This produced high-dimensional sparse feature matrices with over one thousand encoded features in some variants.

3.5 Class Imbalance

The dataset exhibited severe class imbalance. In the complete-case dataset, AML represented only 19.2% of samples. In the reduced-locus dataset, AML represented 26.1%. To address this, two strategies were used: class weighting and SMOTE oversampling. SMOTE produced balanced datasets with equal representation of AML and healthy samples [7].

4 Modeling Approach

4.1 Models Evaluated

A comprehensive suite of classical ML models was evaluated, including logistic regression, support vector classifiers, naive Bayes variants, stochastic gradient descent classifiers, k-nearest neighbors, decision trees, random forests, gradient boosting, bagging, ridge classifiers, and perceptrons. CatBoost was also evaluated on non-encoded datasets due to its native handling of categorical features.

4.2 Hyperparameter Optimization

Grid search was used to tune hyperparameters for each model across all dataset variants. Optimal configurations varied by dataset and model type. For example, logistic regression performed best with strong regularization and class weighting, while gradient boosting required deep trees and many estimators.

4.3 Train-Test Splitting

All datasets were split using an 80/20 stratified split to preserve class proportions. Five-fold cross-validation was used for additional robustness.

5 Results

5.1 Performance on Non-SMOTE Datasets

Models trained on imbalanced datasets consistently failed to detect AML cases. Many models achieved high accuracy by predicting the majority class exclusively. For example, SVC achieved 75% accuracy on the reduced-locus dataset but 0% recall for AML. Logistic regression achieved 62% accuracy but only 21% recall for AML. These results demonstrate that class weighting alone is insufficient in this setting.

5.2 Performance on SMOTE Datasets

SMOTE dramatically improved minority recall. Logistic regression achieved 91% accuracy and 93% recall for AML on the complete-case SMOTE dataset. Gradient boosting and SVC also achieved balanced performance with AML recall above 90%. However, these improvements did not generalize across all dataset variants, and performance remained unstable.

5.3 CatBoost Performance

CatBoost performed moderately well on the reduced-locus dataset, achieving 71% accuracy and 29% recall for AML. Feature importance analysis indicated that age, HLA-A, HLA-B, DRB11, and DRB12 were influential features.

5.4 Permutation Importance

Permutation importance provides a model-agnostic measure of how strongly each feature contributes to predictive performance. By randomly permuting the values of a feature and measuring the resulting drop in model accuracy, this method quantifies the degree to which the model relies on that feature. In the present study, permutation importance was computed on the SMOTE-encoded dataset, where the classifier achieved balanced performance across AML and healthy classes. This makes the resulting importance values more reliable than those obtained from imbalanced datasets.

The plot in Figure 1 shows that several allele-level features exert measurable influence on the model. Notably, the B2.40.0 and B2.51.0 alleles appear among the most influen-

tial features, suggesting that specific HLA-B variants contribute to the model’s decision boundary. DRB5 and DQB12-related alleles also rank highly, indicating that class II HLA variation plays a non-negligible role in shaping the model’s predictions. The presence of sex among the top features reflects demographic differences in the dataset rather than biological causality. Importantly, the magnitude of these importance values remains modest, consistent with the overall weak predictive signal observed in the classification results.

These findings align with prior literature showing that HLA-B and HLA-DRB alleles can influence immune reactivity and NK cell licensing [1,9]. However, the modest effect sizes and lack of consistent predictive power across dataset variants suggest that these associations are insufficient for robust AML classification.

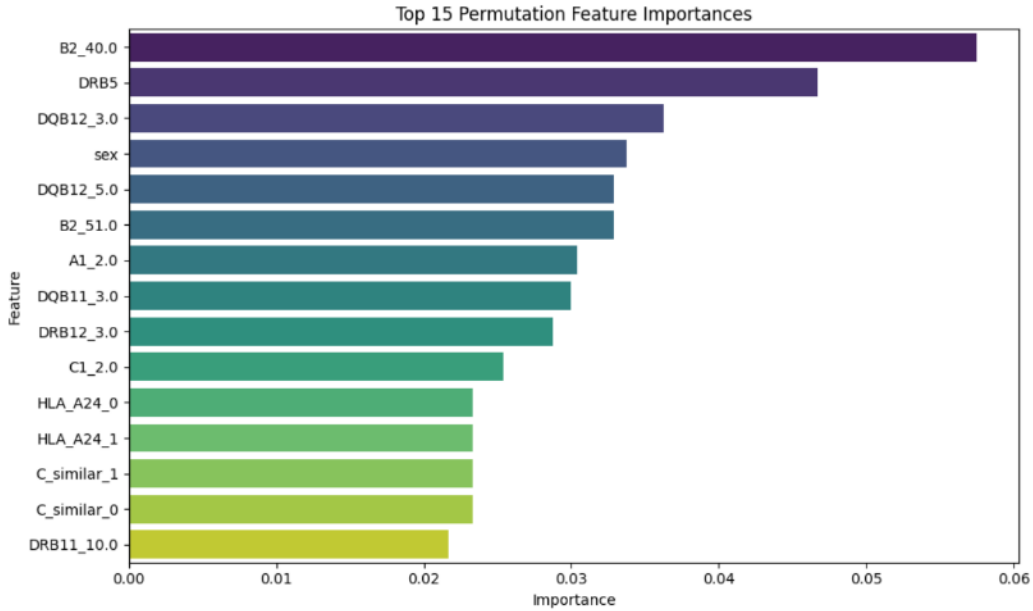


Figure 1: Top 15 Permutation Feature Importances.

5.5 Feature Interaction Analysis

The feature interaction heatmap in Figure 2 visualizes pairwise interaction strengths between features in the reduced-locus dataset. Interaction scores were computed using CatBoosts built-in interaction detection algorithm, which quantifies how much predictive gain arises from considering two features jointly rather than independently. This is particularly relevant for HLA data, where linkage disequilibrium (LD) and haplotype structure create non-linear dependencies between alleles.

The heatmap reveals several biologically plausible interaction patterns. Strong interactions are observed between A1 and A2, B1 and B2, and DRB11 and DRB12, reflecting known LD patterns within HLA class I and class II loci [5]. These interactions mirror the LD matrix computed earlier in the preprocessing stage, confirming that the model captures underlying haplotype structure. A notable interaction between A2 and B1 also appears, which may reflect extended haplotypes spanning class I loci.

Despite these structured interactions, the models predictive performance remains limited. This indicates that while the HLA loci exhibit meaningful genetic correlations, these correlations do not translate into discriminative power for AML classification. This is consistent with prior findings that HLA variation influences immune response and transplant outcomes but does not strongly determine AML susceptibility [2,3].

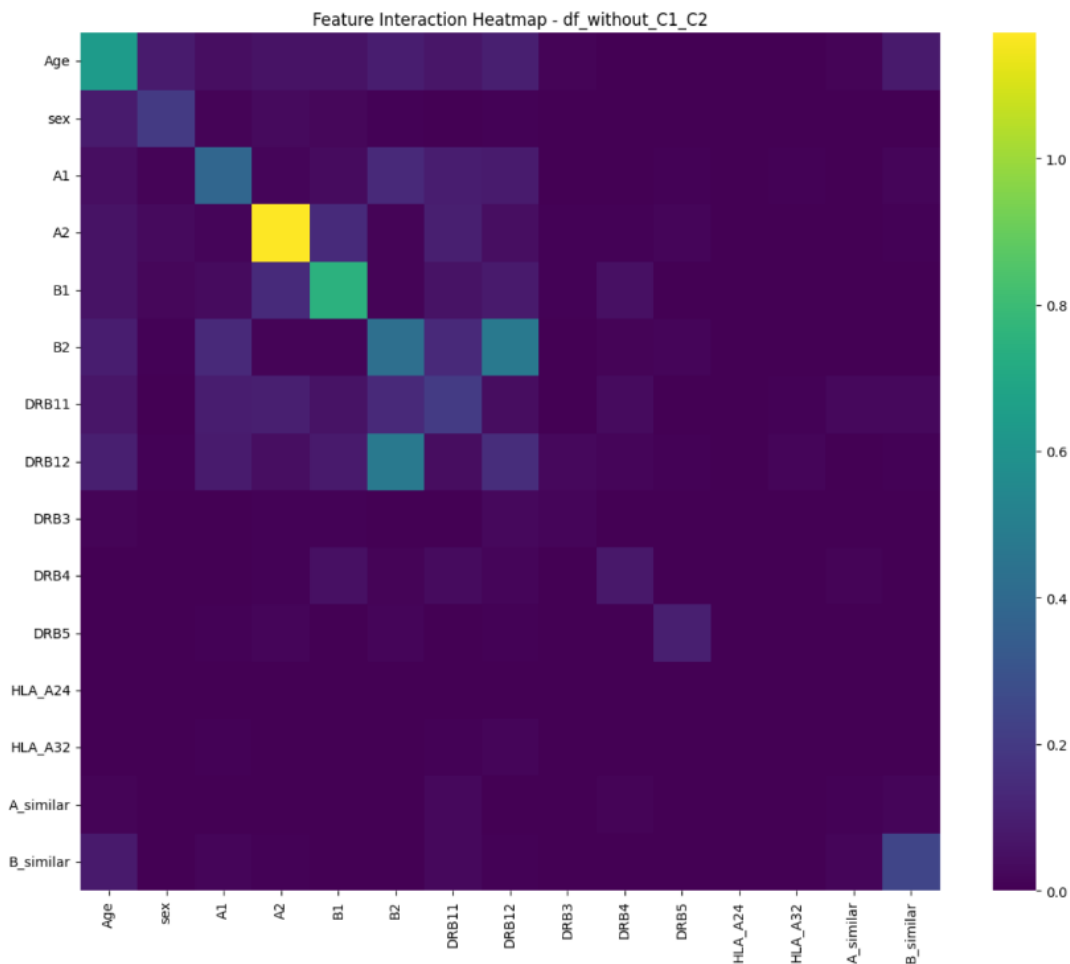


Figure 2: Feature Interaction Heatmap for Reduced-Locus Dataset.

6 Discussion

6.1 Why Models Failed

The primary reason for model failure is the absence of a strong predictive signal in the dataset. HLA genotypes alone are insufficient to distinguish AML from healthy individuals. The extreme sparsity and high dimensionality of allele-level features exacerbate this issue. Class imbalance further biases models toward majority-class predictions. Even with SMOTE, models may overfit synthetic patterns that do not generalize.

6.2 Methodological Insights

This study highlights several methodological lessons. First, classical ML models struggle with sparse immunogenetic data unless sample sizes are large. Second, SMOTE can improve minority recall but may introduce artifacts [7]. Third, feature importance must be interpreted cautiously when models exhibit poor generalization [6]. Fourth, linkage disequilibrium structure influences feature interactions but does not guarantee predictive utility [5].

6.3 Comparison to Literature

While certain HLA alleles have been associated with AML relapse risk, particularly in transplant settings [2], these associations do not translate into predictive power for AML status in the general population. This aligns with broader literature indicating that HLA variation influences immune response rather than disease onset [1, 3, 9].

7 Future Work

Future research should address the methodological constraints identified in this thesis by pursuing several complementary directions. First, larger and more comprehensive datasets are essential for improving model reliability. High-dimensional immunogenetic data require substantial sample sizes to avoid overfitting and to ensure adequate representation of rare alleles. Future studies should prioritize the collection of larger cohorts with high-resolution HLA typing and reduced missingness.

Second, more advanced machine learning models may offer improved performance. Deep learning architectures, such as autoencoders for dimensionality reduction or transformer-based models for sequence-like allele representations, could capture complex non-linear relationships that classical models fail to learn. Graph neural networks may also be suitable for modeling linkage disequilibrium structure and allele co-occurrence patterns.

Third, improved feature engineering and encoding strategies should be explored. Embedding-based representations of HLA alleles, LD-aware encoding schemes, and peptide-binding affinity predictions may provide richer and more biologically meaningful features. These approaches could mitigate the sparsity introduced by one-hot encoding and enhance the models ability to generalize.

Finally, future work should incorporate more robust evaluation strategies, including nested cross-validation and external validation cohorts. These methods would provide a more reliable assessment of model performance and reduce the risk of overfitting to a single dataset.

8 Conclusion

This thesis demonstrates that classical machine learning models cannot reliably predict AML status from HLA genotypes and demographic variables in the available dataset. Despite extensive preprocessing, feature engineering, encoding, class imbalance handling, and model evaluation, no model achieved clinically meaningful performance. The findings underscore the limitations of classical ML in high-dimensional immunogenetic settings and highlight the need for larger, more diverse datasets and integration of advanced modeling techniques.

9 Limitations of the Work

This study is subject to several methodological limitations that constrain the interpretability and generalizability of the results. The dataset used in this thesis was originally collected and curated by a master’s student, Bahare Esmaili Moghadam, under the supervision of Dr. Saeidi and Dr. Ehsan. While the dataset provides a valuable starting point for exploratory machine learning analysis, its structure imposes inherent constraints

on the achievable performance of classical models.

The most significant limitation is the small sample size. With only 294 individuals available prior to preprocessing, the dataset falls well below the scale typically required for reliable machine learning on high-dimensional immunogenetic data. Classical models such as logistic regression, support vector machines, and gradient boosting are sensitive to the ratio of samples to features, and the extreme dimensionality introduced by one-hot encoding exacerbates this imbalance. As a result, even well-regularized models are prone to overfitting and instability.

A second limitation concerns the substantial amount of missing data. Several HLA loci, particularly C1, C2, DQB11, and DQB12, contained extensive null or zero values. Although multiple dataset variants were constructed to mitigate this issue, including complete-case filtering and reduced-locus preprocessing, each approach introduces trade-offs. Removing rows reduces the already limited sample size, while removing loci discards potentially informative features. These structural deficiencies in the dataset limit the capacity of machine learning models to learn consistent patterns.

A third limitation arises from severe class imbalance. AML cases constituted only a minority of the dataset, and although class weighting and SMOTE oversampling were applied, these techniques cannot fully compensate for the lack of genuine minority-class examples. SMOTE, in particular, may introduce synthetic patterns that do not reflect true biological variability, leading to optimistic training performance that does not generalize to unseen data.

Finally, the absence of additional covariates restricts the modeling space. The dataset contains only demographic variables and HLA genotypes, without clinical, molecular, or laboratory features that are known to influence AML risk and progression. From a machine learning perspective, this limits the feature richness necessary for robust classification and places disproportionate weight on sparse allele-level features.

Taken together, these limitations indicate that the negative results reported in this thesis should not be interpreted as evidence against the existence of immunogenetic patterns in AML. Rather, they reflect the constraints imposed by the dataset’s size, sparsity, and imbalance. The methodological pipeline developed in this work remains valid and can be applied to larger, more comprehensive datasets in future research.

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